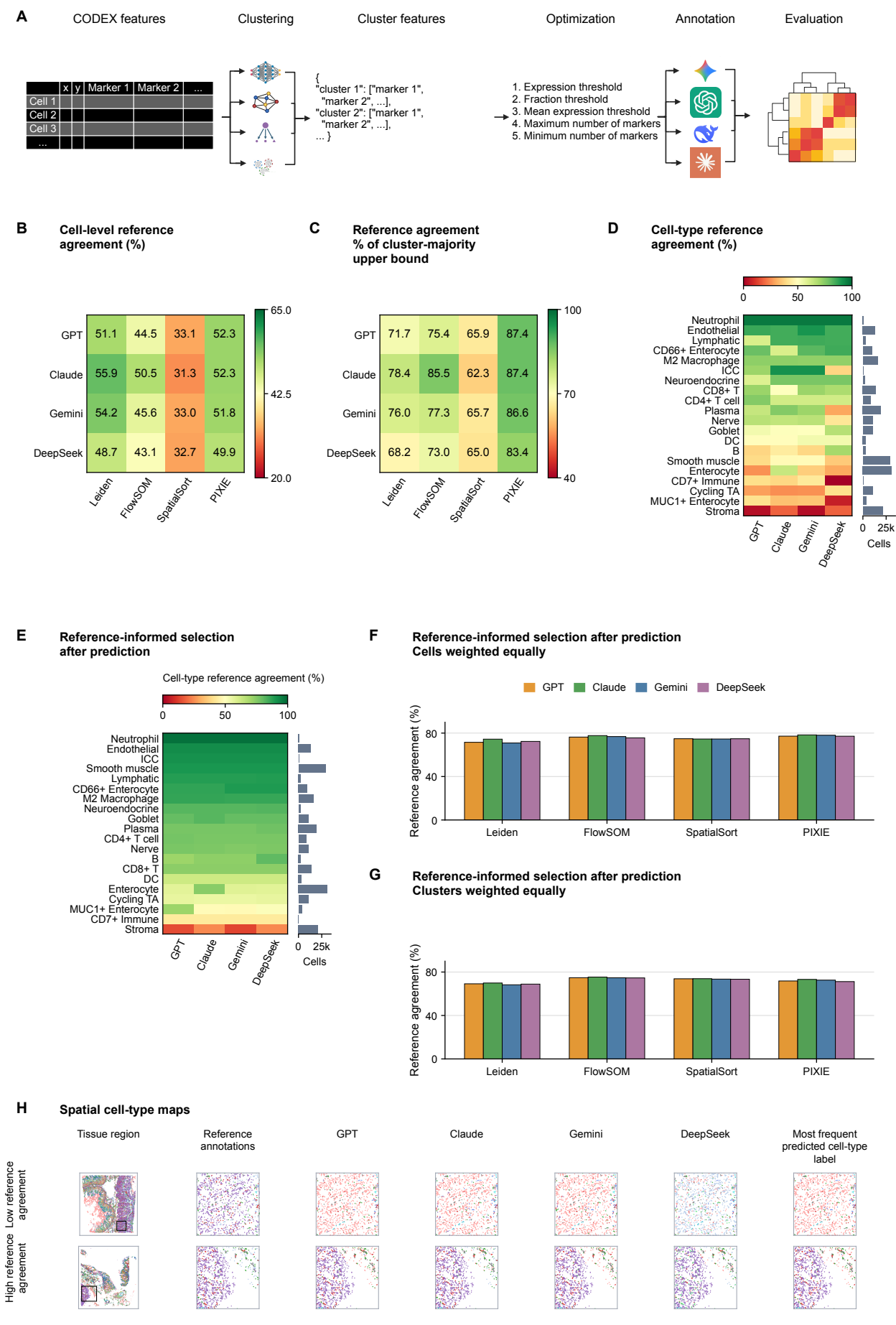

A
CODEX features
Clustering
Cluster features
Optimization
Annotation
Evaluation
x
y
Marker 1
Marker 2
...
Cell 1
Cell 2
Cell 3
...
{
"cluster 1": ["marker 1",
"marker 2", ...],
"cluster 2": ["marker 1",
"marker 2", ...],
... }
1. Expression threshold
2. Fraction threshold
3. Mean expression threshold
4. Maximum number of markers
5. Minimum number of markers
B
Cell-level reference agreement (%)
GPT 51.1 44.5 33.1 52.3
Claude 55.9 50.5 31.3 52.3
Gemini 54.2 45.6 33.0 51.8
DeepSeek 48.7 43.1 32.7 49.9
65.0
42.5
20.0
Leiden
FlowSOM
SpatialSort
PIXIE
C
Reference agreement % of cluster-majority upper bound
GPT 71.7 75.4 65.9 87.4
Claude 78.4 85.5 62.3 87.4
Gemini 76.0 77.3 65.7 86.6
DeepSeek 68.2 73.0 65.0 83.4
100
70
40
Leiden
FlowSOM
SpatialSort
PIXIE
D
Cell-type reference agreement (%)
0
50
100
Neutrophil
Endothelial
Lymphatic
CD66+ Enterocyte
M2 Macrophage
ICC
Neuroendocrine
CD8+ T
CD4+ T cell
Plasma
Nerve
Goblet
DC
B
Smooth muscle
Enterocyte
CD7+ Immune
Cycling TA
MUC1+ Enterocyte
Stroma
GPT
Claude
Gemini
DeepSeek
0
25k
Cells
E
Reference-informed selection after prediction
Cell-type reference agreement (%)
0
50
100
Neutrophil
Endothelial
ICC
Smooth muscle
Lymphatic
CD66+ Enterocyte
M2 Macrophage
Neuroendocrine
Goblet
Plasma
CD4+ T cell
Nerve
B
CD8+ T
DC
Enterocyte
Cycling TA
MUC1+ Enterocyte
CD7+ Immune
Stroma
GPT
Claude
Gemini
DeepSeek
0
25k
Cells
F
Reference-informed selection after prediction
Cells weighted equally
GPT
Claude
Gemini
DeepSeek
Reference agreement (%)
80
40
0
Leiden
FlowSOM
SpatialSort
PIXIE
G
Reference-informed selection after prediction
Clusters weighted equally
Reference agreement (%)
80
40
0
Leiden
FlowSOM
SpatialSort
PIXIE
H
Spatial cell-type maps
Tissue region
Reference annotations
GPT
Claude
Gemini
DeepSeek
Most frequent predicted cell-type label
Low reference agreement
High reference agreement